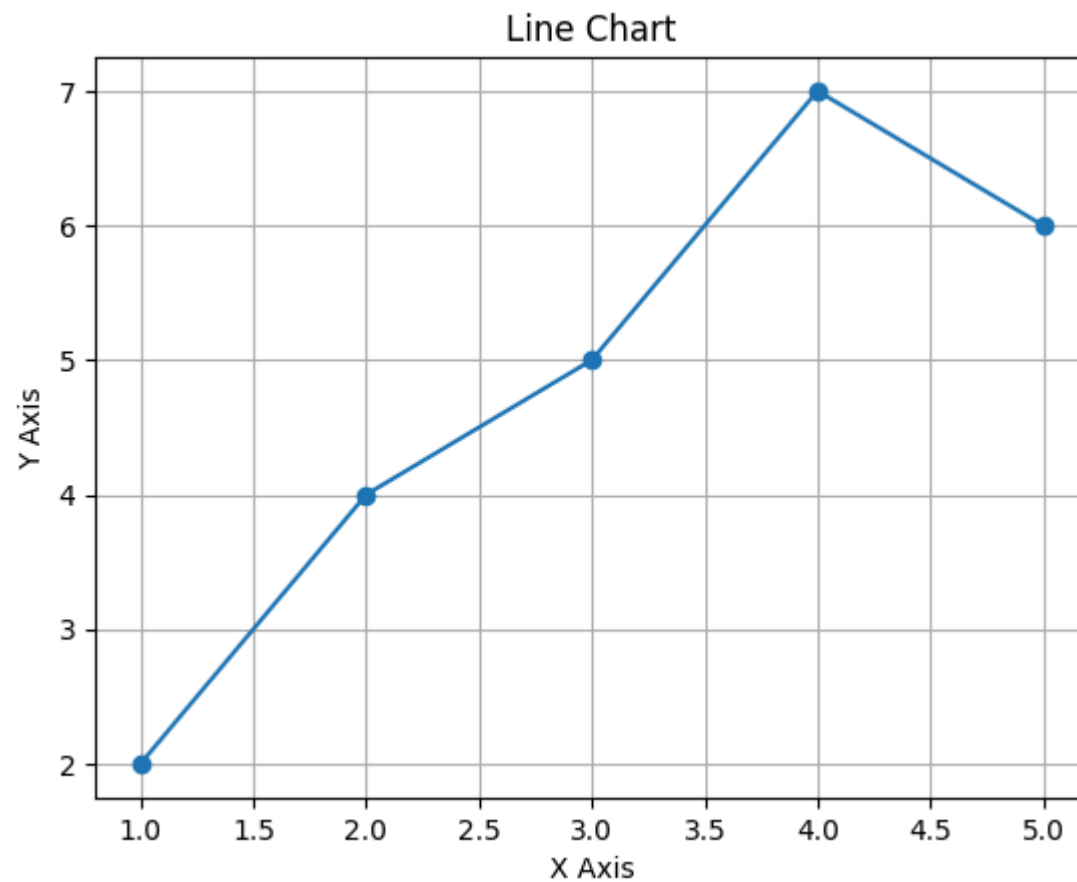


```
In [7]: import matplotlib.pyplot as plt
x = [1,2,3,4,5]
y = [2,4,5,7,6]
```

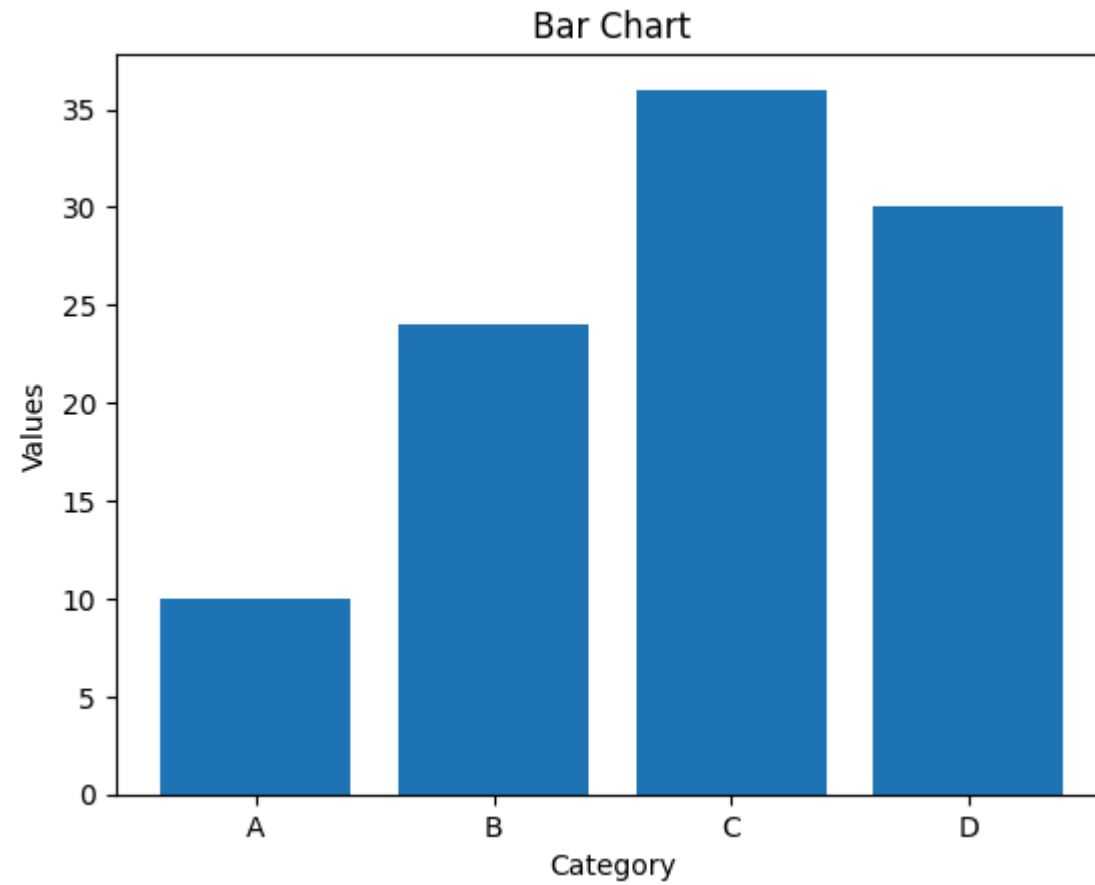
```
In [8]: #Line Chart
plt.plot(x,y, marker='o')
plt.title("Line Chart")
plt.xlabel("X Axis")
plt.ylabel("Y Axis")
plt.grid(True)
plt.show()
```



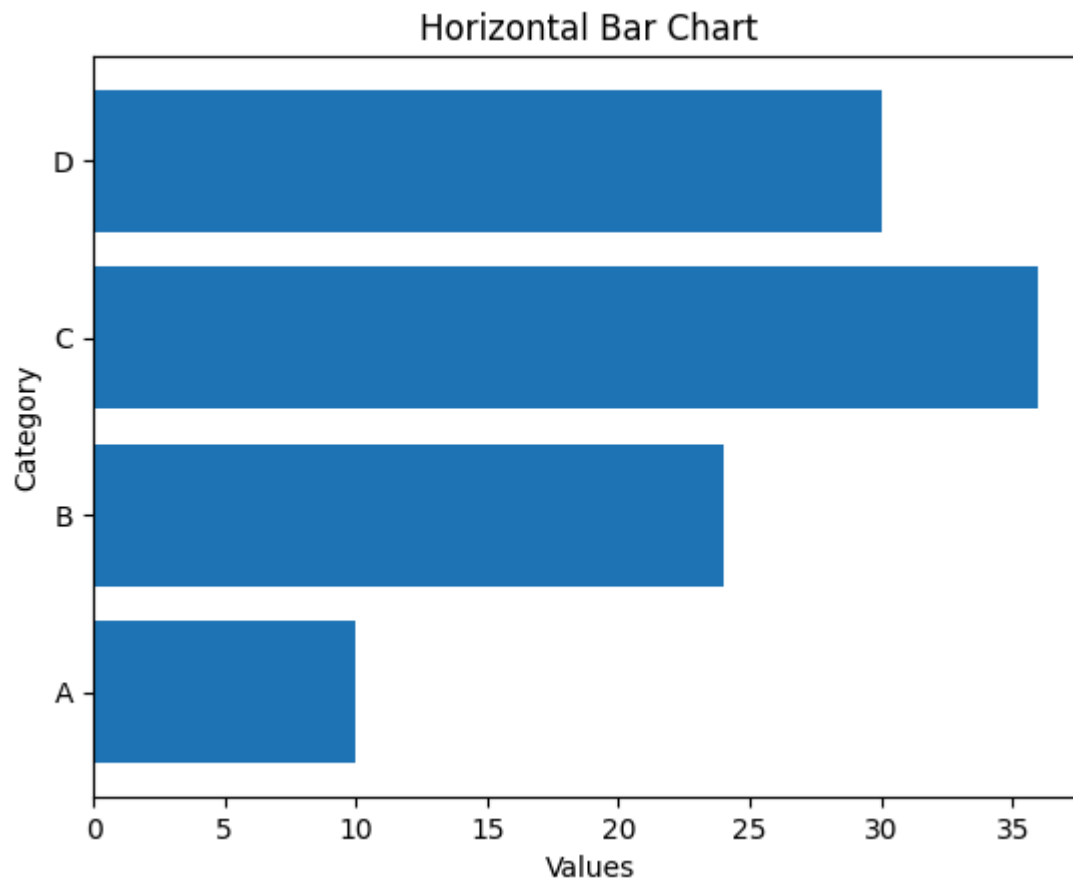
```
In [11]: #Bar Chart

Categories = ['A', 'B', 'C', 'D']
Values = [10, 24, 36, 30]

plt.bar(Categories, Values)
plt.title("Bar Chart")
plt.xlabel("Category")
plt.ylabel("Values")
plt.show()
```



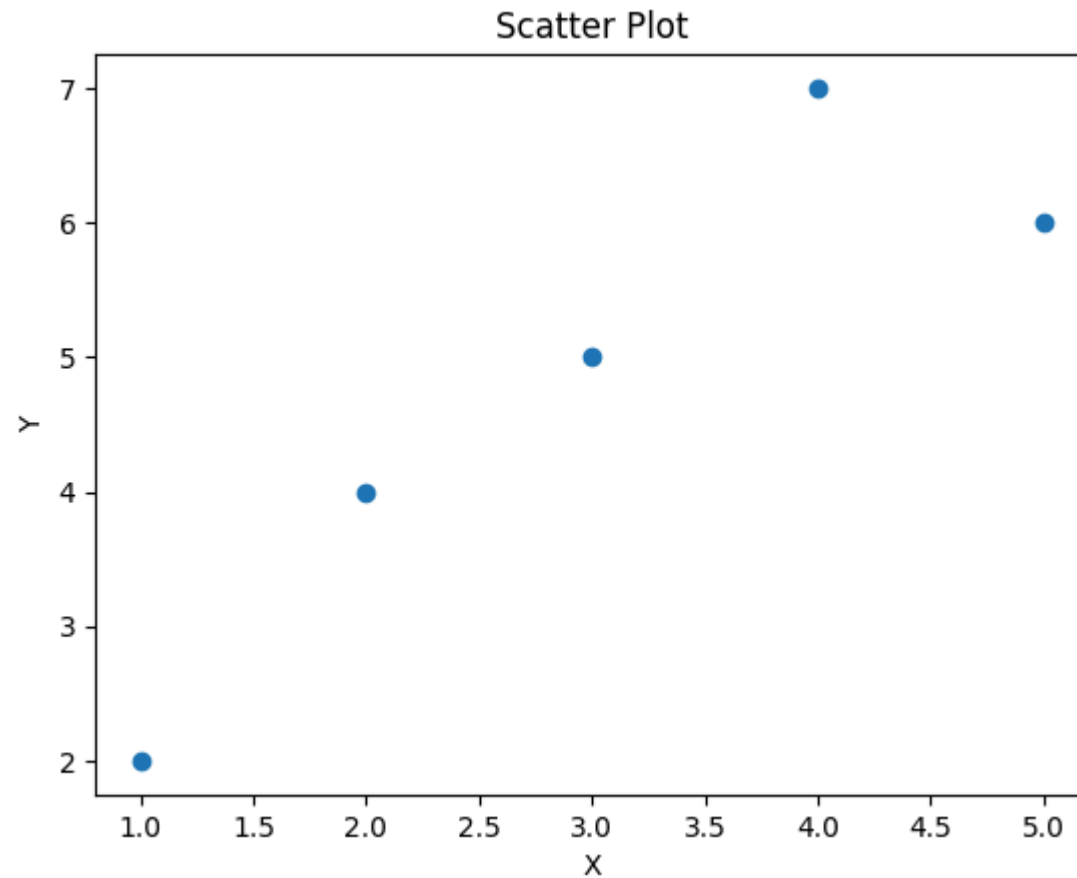
```
In [12]: #Horizontal Bar Chart
plt.barh(Categories, Values)
plt.title("Horizontal Bar Chart")
plt.xlabel("Values")
plt.ylabel("Category")
plt.show()
```



```
In [13]: #Scatter Plot

scatterX = [5,7,8,7,2,17,2,9]
scatterY = [99,86,87,88,100,86,103,87]
```

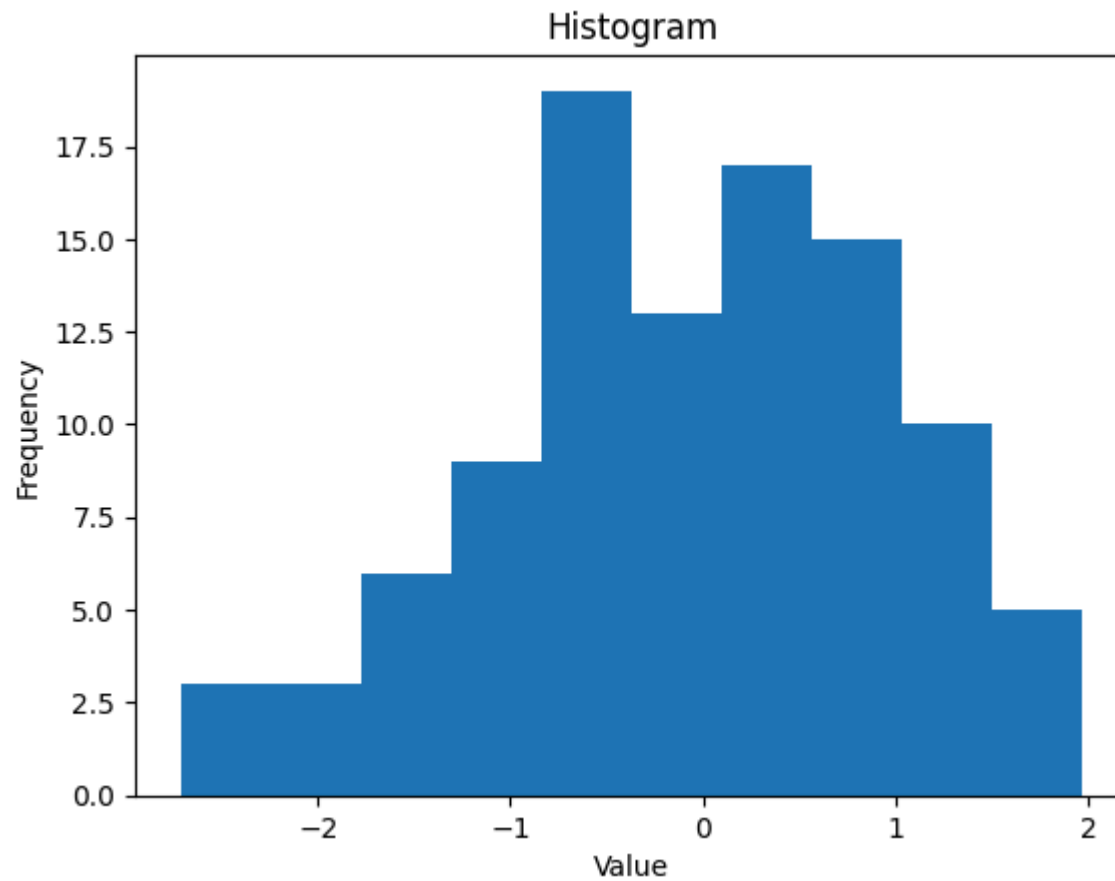
```
plt.scatter(x,y)
plt.title("Scatter Plot")
plt.xlabel("X")
plt.ylabel("Y")
plt.show()
```



```
In [19]: #Histogram
import numpy as np
data = np.random.randn(100)

plt.hist(data, bins = 10)
plt.title("Histogram")
plt.xlabel("Value")
```

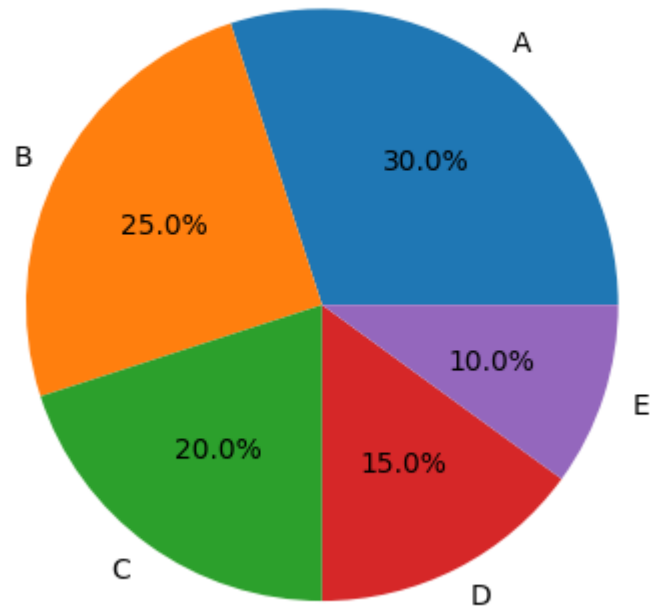
```
plt.ylabel("Frequency")  
plt.show()
```



```
In [31]: #Pie chart  
sizes = [30,25,20,15,10]  
labels = ['A', 'B', 'C', 'D', 'E']  
  
plt.pie(sizes, labels = labels, autopct='%1.1f%%')  
plt.title("Pie Chart")  
plt.show
```

```
Out[31]: <function matplotlib.pyplot.show(close=None, block=None)>
```

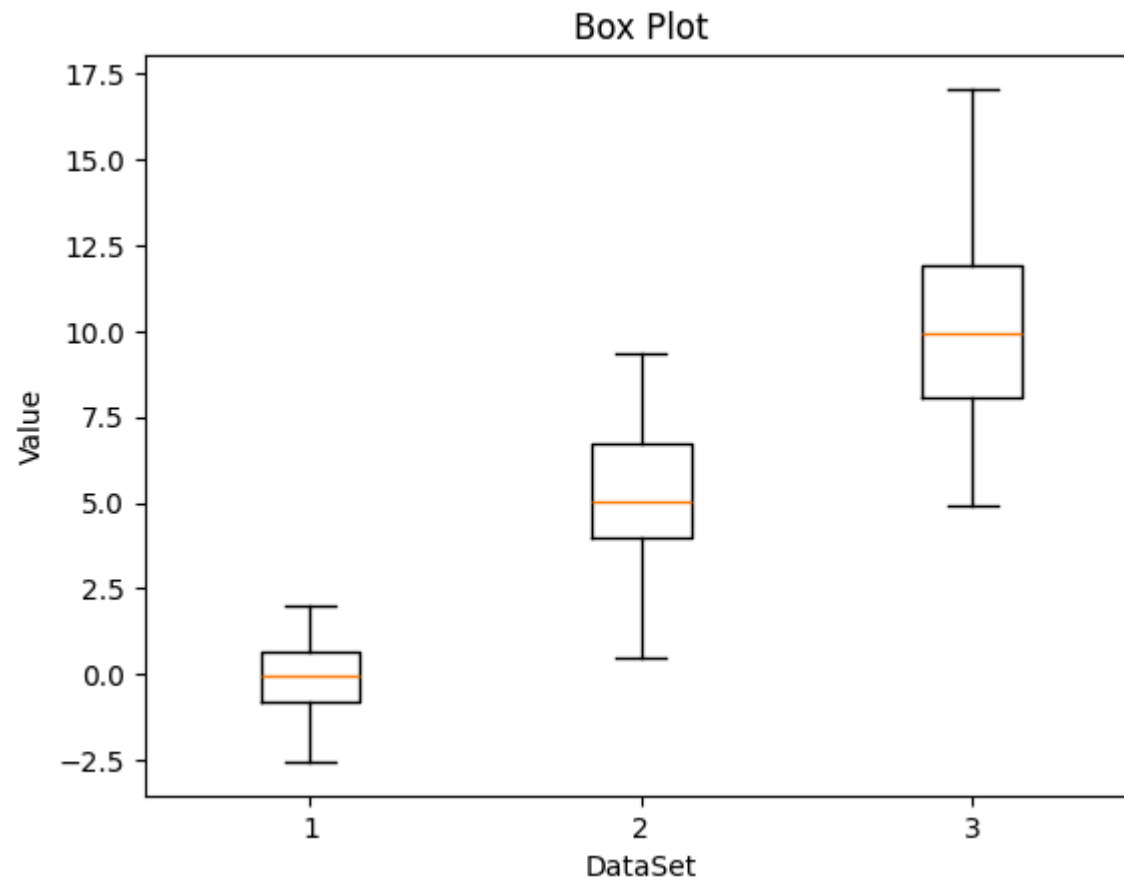
Pie Chart



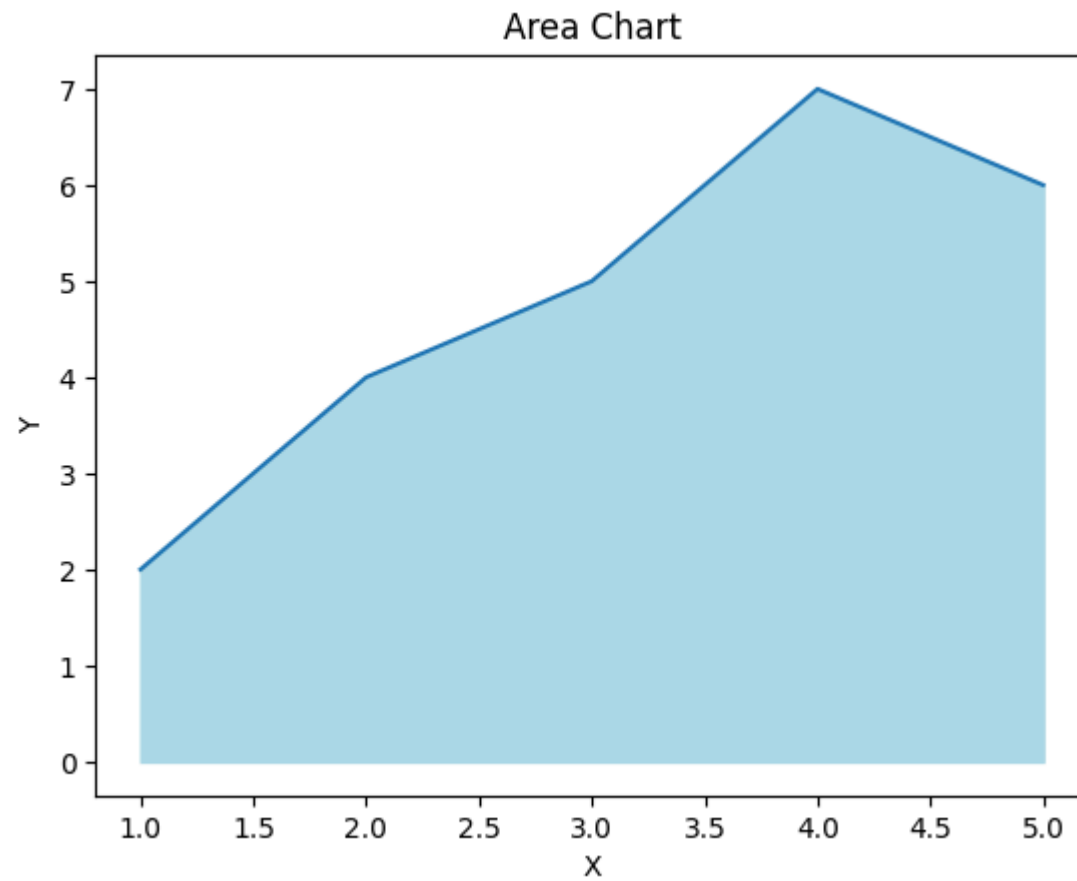
In [33]: *# Box Plot*

```
data = [np.random.normal(0,1,100), (np.random.normal(5,2,100)), np.random.normal(10,3,100)]

plt.boxplot(data)
plt.title("Box Plot")
plt.xlabel("DataSet")
plt.ylabel("Value")
plt.show()
```



```
In [36]: # Area Chart
plt.fill_between(x,y, color='lightblue')
plt.plot(x,y)
plt.title("Area Chart")
plt.xlabel("X")
plt.ylabel("Y")
plt.show()
```



```
In [37]: # Multiple Line Chart

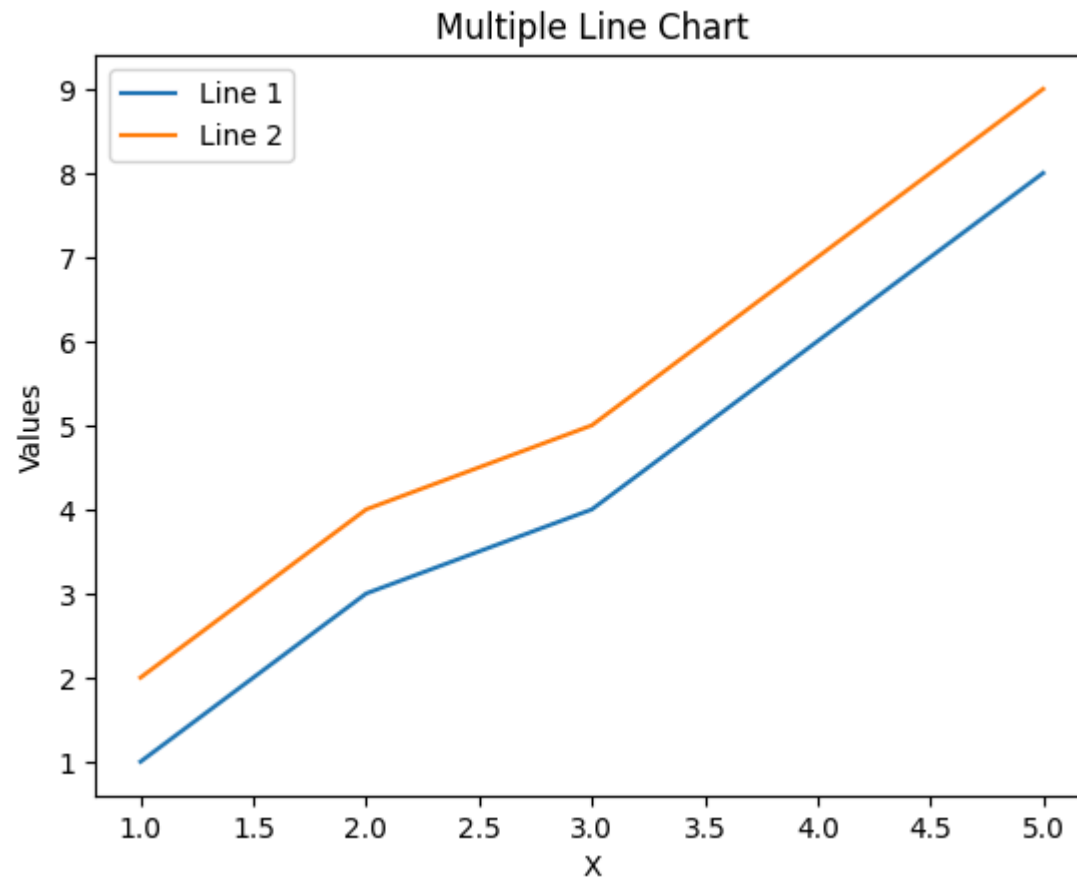
y1=[1,3,4,6,8]
y2=[2,4,5,7,9]

plt.plot(x, y1, label = "Line 1")
plt.plot(x, y2, label="Line 2")

plt.title("Multiple Line Chart")
plt.legend()
plt.xlabel("X")
```



```
plt.ylabel("Values")  
plt.show()
```



```
In [39]: # Heat Map  
  
data = np.random.rand(5,5)  
print(data)  
  
plt.imshow(data, cmap='viridis')  
plt.colorbar()  
plt.title("HeatMap")  
plt.show()
```

```
[[0.59167652 0.66191602 0.93244081 0.39005433 0.2271778 ]  
 [0.62996609 0.74534768 0.66196164 0.02771099 0.52576594]  
 [0.93248072 0.17245478 0.90210032 0.81704371 0.17082207]  
 [0.09573826 0.94643347 0.4210455  0.37581588 0.66045886]  
 [0.35240524 0.24904649 0.32255175 0.39757735 0.23413475]]
```

